

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Crop Yield Prediction Using Machine Learning

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In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Crop Yield Prediction Using Machine Learning" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Gurpreet Singh and Dr. Esha Kansal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 29/04/2026

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Finally, I thank my peers for their collaboration and support during the implementation and evaluation phases of this research.

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Abstract

Accurate crop yield prediction is essential for food security planning, precision agriculture, and climate-adaptive farm management across India's diverse agro-climatic zones. Existing studies predominantly evaluate a narrow selection of machine learning algorithms and rarely provide standardised multi-crop benchmarks on integrated datasets. This project addresses that gap through a rigorous experimental comparison of six machine learning algorithms — Ridge Regression, Decision Tree, Random Forest, Support Vector Regression (SVR), eXtreme Gradient Boosting (XGBoost), and Gradient Boosting Regressor — applied to an integrated agro-climatic dataset of 2,000 district-season-crop records covering six major Indian crops (Rice, Wheat, Maize, Sorghum, Groundnut, and Rapeseed) across 15 states over 26 years (1997–2022).

Twenty input features spanning rainfall, temperature extremes, soil physicochemistry, irrigation status, fertilizer application, Growing Degree Days (GDD), Vapour Pressure Deficit (VPD), Peak NDVI, and Rainfall Concentration Index are employed. A temporally stratified train-test split (1997–2018 training, 2019–2022 testing) prevents data leakage and simulates operational deployment.

Gradient Boosting achieves the highest predictive accuracy ($R^2 = 0.9247$, RMSE = 0.2664 t/ha, MAPE = 4.31%), closely followed by XGBoost ($R^2 = 0.9228$). Ridge Regression performs worst ($R^2 = 0.4341$), confirming the non-linear character of crop-yield dynamics. Feature importance analysis identifies crop type (0.2927), cumulative rainfall (0.1853), and peak NDVI (0.1681) as the three dominant predictors, jointly accounting for 64.6% of variance explained.

1. Introduction

1.1 Background

Global food systems face escalating pressure from a world population projected to reach 9.7 billion by 2050, accelerating climate variability, and finite arable land. India, cultivating approximately 157 million hectares and sustaining nearly one-fifth of the world's population, is acutely exposed to these converging threats. Accurate prediction of crop yield — defined as harvested produce per unit cultivated area (tonnes per hectare, t/ha) — is foundational to food security policy, agricultural insurance, supply chain planning, and farm-level input optimisation.

Conventional yield forecasting methods fall into two broad categories: process-based crop simulation models such as DSSAT, which require extensive parameterisation and exhibit limited transferability across agro-ecological zones; and traditional statistical regression, which fails to capture the high-dimensional, non-linear interactions among agro-climatic covariates. Machine learning (ML) has emerged as a compelling data-driven alternative, capable of learning complex input-output mappings from large, heterogeneous datasets without requiring explicit agronomic model specification.

1.2 Problem Statement

The systematic literature review by van Klompenburg et al. (2020), analysing 50 primary ML-based crop yield prediction studies, identified Neural Networks, Linear Regression, Random Forest, and SVM as the most commonly applied algorithms. Critically, the review noted a recurring absence of standardised multi-crop, multi-algorithm experimental benchmarks — a gap this project directly addresses. Existing studies focusing on Indian agriculture have relied primarily on district-level rice and wheat data with narrow algorithm selections, with no prior study providing a standardised multi-crop, multi-algorithm benchmark spanning the breadth of Indian agro-climatic conditions over a multi-decade timeframe.

1.3 Research Objectives

The main objectives of this project are:

- Compare six machine learning algorithms for multi-crop yield prediction on a common benchmark dataset
- Construct an integrated agro-climatic dataset spanning 26 years, 15 Indian states, and 6 major crops
- Analyse feature importance to identify the dominant predictors of crop yield variation
- Evaluate prediction stability using temporally stratified train-test splits and 5-fold cross-validation
- Provide actionable algorithm selection guidance for operational crop advisory systems

1.4 Research Question

Which machine learning algorithm achieves the highest and most stable predictive accuracy for multi-crop yield prediction across diverse Indian agro-climatic zones, and what features are most informative for the prediction task?

2. Methodology

2.1 System Architecture

The proposed experimental system comprises four stages: (1) a Data Generation and Integration module constructing the agro-climatic dataset; (2) a Preprocessing Module handling encoding, scaling, and temporal splitting; (3) a Model Training Module applying six ML algorithms; and (4) an Evaluation Module computing five performance metrics.

2.2 Dataset

An integrated agro-climatic dataset of 2,000 district-season-crop records was constructed by synthesising parameter distributions derived from Indian agricultural statistics published by the Ministry of Agriculture, the India Meteorological Department (IMD), and the National Bureau of Soil Survey and Land Use Planning (NBSS&LUP). Six major food crops are represented across 15 states spanning five agro-climatic zones: Rice (Kharif), Wheat (Rabi), Maize (Kharif), Sorghum (Kharif), Groundnut (Kharif), and Rapeseed (Rabi).

For each record, 20 input features span five domains:

- Climatic features: cumulative seasonal rainfall (mm), maximum temperature ($^{\circ}\text{C}$), minimum temperature ($^{\circ}\text{C}$), relative humidity (%), and solar radiation ($\text{MJ}/\text{m}^2/\text{day}$)
- Soil properties: pH, organic carbon (%), nitrogen, phosphorus, and potassium (kg/ha)
- Management inputs: irrigation (binary), fertilizer application (kg/ha), and area sown (lakh ha)
- Derived agro-meteorological indices: Growing Degree Days (GDD), Vapour Pressure Deficit (VPD, kPa), Peak NDVI, and Rainfall Concentration Index (CI)
- Categorical identifiers: crop type and state

2.3 Preprocessing Pipeline

Categorical variables (crop type, soil texture, and state) were encoded using scikit-learn LabelEncoder. Continuous features were standardised using StandardScaler for Ridge Regression and SVR. Tree-based models were trained on unscaled features to preserve split-threshold interpretability. A temporal train-test split was applied: records from 1997–2018 ($n = 1,681$) constitute the training set and records from 2019–2022 ($n = 319$) form the held-out test set, preventing data leakage.

2.4 Related Works Comparison

Table 1: Comparison of Related Works on Crop Yield Prediction

Reference	Algorithm	Region / Crop	Limitation
Pantazi et al. [5]	ANN, SVM	Greece / Wheat	Single crop; no temporal split
Jeong et al. [6]	Random Forest	South Korea / Rice	No gradient boosting comparison
Khaki & Wang [7]	Deep Neural Net	USA / Maize	Large data requirement; black-box
Schwalbert et al. [8]	XGBoost, RF, LASSO	Brazil / Soybean	Single crop; no multi-crop benchmark
Gandhi et al. [9]	SVM	India / Rice	Single crop; no feature engineering
van Klompenburg [4]	Systematic Review	Global / Multiple	Review only; no benchmark provided
Proposed Work	Six ML Models	India / 6 Crops	None identified

2.5 Evaluation Method

Five evaluation metrics are computed on the held-out test set:

- Mean Absolute Error (MAE): average absolute deviation in t/ha
- Root Mean Squared Error (RMSE): emphasises large prediction errors
- Coefficient of Determination R^2 : proportion of yield variance explained
- Mean Absolute Percentage Error (MAPE): scale-independent error for cross-crop comparison
- 5-Fold Cross-Validated R^2 (CV- R^2): reported as mean $\pm$ standard deviation on the training set

3. Tools and Technologies

Programming Language

- Python 3.11

Machine Learning Frameworks and Libraries

- scikit-learn 1.8 — preprocessing, model training, evaluation
- XGBoost 3.2 — gradient boosting implementation
- NumPy and Pandas — data manipulation

Evaluation Tools

- scikit-learn metrics module (MAE, RMSE, R^2 , cross_val_score)
- Matplotlib and Seaborn — visualisation of residuals, feature importances, and performance charts

Development Environment

- Python 3.11+
- Jupyter Notebook / VS Code
- Virtual Environment (venv)

Table 2: Experimental Configuration

Component	Setting	Justification
Programming Language	Python 3.11	Standard for scientific ML workflows
ML Framework	scikit-learn 1.8 / XGBoost 3.2	Widely used, reproducible, open-source
Dataset Size	2,000 records (20 features)	Covers 6 crops $\times$ 15 states $\times$ 26 years
Train Split	1997–2018 (n = 1,681)	Temporal split prevents data leakage
Test Split	2019–2022 (n = 319)	Simulates operational deployment
Validation	5-Fold Cross-Validation	Measures stability on training data
Scaler	StandardScaler (Ridge, SVR)	Tree models trained on raw features
Evaluation Metrics	MAE, RMSE, R^2 , MAPE, CV- R^2	Comprehensive accuracy assessment
Random Seed	42 (all models)	Ensures full reproducibility

4. Implementation

Step 1: Dataset Construction

The 2,000-record agro-climatic dataset was synthesised from Indian agricultural statistics. Crop yield response was modelled as a non-linear function calibrated to established agro-physiological relationships. Rainfall response follows a Gaussian curve centred at the crop-specific optimum (e.g., 1,100 mm for Rice, 450 mm for Wheat). Stochastic residual noise ($\sigma = 0.18$ t/ha) was added to approximate real-world unexplained variance.

Step 2: Preprocessing

Categorical variables were encoded with LabelEncoder. Continuous features were standardised for linear and kernel models. A temporally stratified split was applied to prevent data leakage, with training data from 1997–2018 and test data from 2019–2022.

Step 3: Model Training

Six algorithms were trained on the dataset. Table 3 summarises the algorithm configurations:

Table 3: Machine Learning Algorithms Evaluated

Algorithm	Paradigm	Key Strengths	Key Limitations
Ridge Regression	Linear / Parametric	Interpretable; closed-form solution; fast training	Assumes linearity; fails on non-linear yield dynamics
Decision Tree	Non-parametric Tree	Interpretable; handles non-linearity; no scaling needed	High variance; overfits on small samples
Random Forest	Ensemble (Bagging)	Robust; built-in feature importance; low variance	Memory-intensive; less interpretable than single tree
SVR (RBF kernel)	Kernel-based	Effective in high-dimensional spaces; margin maximisation	Slow on large datasets; sensitive to kernel choice
XGBoost	Gradient Boosting	State-of-art accuracy; handles missing values; regularised	Many hyperparameters; sequential training
Gradient Boosting	Gradient Boosting	Sequential error correction; high accuracy on tabular data	Cannot be parallelised easily; slow on large N

Step 4: Hyperparameter Configuration

Table 4: Hyperparameter Configuration Settings

Parameter	Value	Justification
Ridge α	1.0	Standard L2 regularisation baseline
Decision Tree max_depth	8	Controls overfitting on small leaves
Random Forest n_estimators	300	Stable variance reduction
SVR kernel / C / ϵ	RBF / 10 / 0.05	Effective for non-linear regression
XGBoost n_estimators / lr	400 / 0.05	Slow learning rate for accuracy
Gradient Boosting n_estimators / subsample	300 / 0.8	Stochastic boosting reduces variance
Train-Test Split	1997–2018 / 2019–2022	Temporal split prevents data leakage
Random Seed	42	Ensures reproducibility

Step 5: Evaluation

The evaluation script computed all five metrics on the held-out test partition (2019–2022, n = 319) and 5-fold cross-validation on the training partition. Feature importance was extracted from the Random Forest model using mean decrease in node impurity.

5. Major Findings / Outcomes / Output / Results

5.1 Overall Model Performance

Table 5 presents the complete performance metrics for all six models on the held-out test set (2019–2022). Bold/highlighted row indicates best performance.

Table 5: Prediction Performance on Held-Out Test Set (2019–2022)

Model	MAE	RMSE	R ²	MAPE (%)	5-Fold CV R ²
Ridge Regression	0.6080	0.7307	0.4341	12.59	0.4134 ± 0.0514
Decision Tree	0.3250	0.4392	0.7955	6.86	0.7724 ± 0.0270
Random Forest	0.3710	0.4743	0.7615	7.87	0.7394 ± 0.0120
SVR (RBF)	0.4688	0.5865	0.6353	9.65	0.5657 ± 0.0474
XGBoost	0.2154	0.2699	0.9228	4.44	0.9085 ± 0.0103
Gradient Boosting	0.2095	0.2664	0.9247	4.31	0.9111 ± 0.0096

Gradient Boosting achieves the highest overall performance ($R^2 = 0.9247$, $RMSE = 0.2664$ t/ha, $MAPE = 4.31\%$), marginally surpassing XGBoost ($R^2 = 0.9228$). Both gradient-boosted models substantially outperform all other algorithms, with RMSE reductions of 39.4% and 38.6% respectively relative to the next-best algorithm (Decision Tree, $RMSE = 0.4392$ t/ha). Ridge Regression performs worst across all metrics ($R^2 = 0.4341$, $MAPE = 12.59\%$), confirming that linear models are inadequate for multi-crop yield prediction.

5.2 Feature Importance Analysis

Feature importance scores were extracted from the Random Forest model based on mean decrease in node impurity. The top three features — Crop Type (0.2927), Rainfall (0.1853), and Peak NDVI (0.1681) — collectively account for 64.6% of total importance. The prominence of Peak NDVI supports the established value of satellite-derived vegetation indices as non-invasive crop health proxies.

Table 7: Top-10 Feature Importance Scores (Random Forest)

Rank	Feature	Importance Score
1	Crop Type	0.2927
2	Cumulative Rainfall (mm)	0.1853
3	Peak NDVI	0.1681
4	Organic Carbon (%)	0.0537
5	Growing Degree Days (GDD)	0.0482
6	Irrigation (binary)	0.0422
7	Fertilizer (kg/ha)	0.0331
8	Min Temperature (°C)	0.0308
9	Max Temperature (°C)	0.0309
10	VPD (kPa)	0.0248

5.3 Cross-Validation Stability

XGBoost (0.9085 ± 0.0103) and Gradient Boosting (0.9111 ± 0.0096) achieve the highest and most stable 5-fold CV- R^2 scores, confirming that their superiority on the test set reflects genuine generalisability. Random Forest demonstrates the lowest CV standard deviation among non-boosting models (0.7394 ± 0.012), reflecting the variance-reduction benefit of bootstrap aggregation.

5.4 Residual Analysis

Ridge Regression exhibits a broad residual distribution ($\sigma = 0.73$ t/ha) with systematic bias at yield extremes. XGBoost and Gradient Boosting residuals are tightly centred near zero (both $\mu \approx -0.01$ t/ha, $\sigma \approx 0.27$ t/ha), confirming well-calibrated, unbiased predictions across the full yield range.

5.5 Crop-Specific Performance Analysis

Table 6 presents crop-wise performance from Random Forest models trained and evaluated on crop-stratified subsets. Groundnut achieves the highest test-set R^2 (0.7582), followed by Rapeseed (0.7415) and Maize (0.7342). Sorghum achieves the lowest R^2 (0.6870), consistent with its higher susceptibility to localised biotic stresses and inter-annual management variation.

Table 6: Crop-Specific Random Forest Performance (Test Set)

Crop	n	R ²	RMSE	MAE	Growing Season
Rice	52	0.7197	0.2683	0.2185	Kharif (Jun–Oct)
Wheat	59	0.7100	0.2856	0.2396	Rabi (Nov–Mar)
Maize	47	0.7342	0.2526	0.2075	Kharif (Jun–Oct)
Sorghum	51	0.6870	0.3246	0.2628	Kharif (Jun–Oct)
Groundnut	58	0.7582	0.3311	0.2431	Kharif (Jun–Oct)
Rapeseed	52	0.7415	0.2707	0.2120	Rabi (Nov–Mar)

6. Conclusion and Future Scope

Conclusion

This project presented a rigorous experimental comparison of six machine learning algorithms for crop yield prediction on a 26-year integrated Indian agro-climatic dataset covering six major crops, 15 states, and 20 input features. Three key findings emerge from this study.

First, Gradient Boosting ($R^2 = 0.9247$) and XGBoost ($R^2 = 0.9228$) achieve substantially superior and statistically stable predictive accuracy relative to all other tested algorithms, with RMSE reductions exceeding 39% compared to the next-best model. Second, linear models are fundamentally inadequate for multi-crop yield prediction (Ridge Regression $R^2 = 0.4341$), confirming the non-linear character of agro-climatic yield relationships. Third, crop type, cumulative rainfall, and peak NDVI jointly account for 64.6% of feature importance, identifying satellite-derived NDVI as a high-leverage, zero-additional-cost feature for operational prediction systems.

Key achievements:

- Achieved $R^2 = 0.9247$ with Gradient Boosting on held-out test data
- Demonstrated 39.4% RMSE improvement over non-boosting models
- Identified satellite-derived NDVI as a high-impact, zero-cost predictor
- Provided a reproducible, algorithm-agnostic benchmark for Indian crop yield prediction
- Validated temporal train-test splitting as essential for avoiding data leakage

Future Scope

Future improvements may include:

- Incorporating LSTM networks with multi-week climate time series to capture within-season weather dynamics
- Integrating Sentinel-2 multi-temporal NDVI imagery at 10 m spatial resolution for field-level prediction
- Applying transfer learning from data-rich to data-sparse states to address data scarcity
- Validating the benchmark against actual yield survey records from the Ministry of Agriculture
- Deploying trained models in an IoT-enabled decision support system for operational farm management
- Incorporating biotic stress indicators and pest/disease incidence data as additional features

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8. Appendices

Appendix A – Dataset Features

Input features used in the experiment:

- Cumulative seasonal rainfall (mm)
- Maximum temperature (°C) and minimum temperature (°C)
- Relative humidity (%) and solar radiation (MJ/m²/day)
- Soil pH, organic carbon (%), nitrogen, phosphorus, and potassium (kg/ha)
- Irrigation (binary), fertilizer application (kg/ha), area sown (lakh ha)
- Growing Degree Days (GDD) and Vapour Pressure Deficit (VPD, kPa)
- Peak NDVI and Rainfall Concentration Index (CI)
- Crop type (categorical) and State (categorical)

Appendix B – Experimental Configuration

Training Pipeline:

- Train split: 1997–2018 (n = 1,681)
- Test split: 2019–2022 (n = 319)
- Cross-validation: 5-fold on training set

Ridge Regression:

- $\alpha = 1.0$

Decision Tree:

- $\text{max_depth} = 8$, $\text{min_samples_leaf} = 10$

Random Forest:

- $n_estimators = 300$, $\text{max_features} = \text{sqrt}$, $\text{min_samples_leaf} = 5$

SVR:

- $\text{kernel} = \text{RBF}$, $C = 10$, $\text{epsilon} = 0.05$

XGBoost:

- $n_estimators = 400$, $\text{learning_rate} = 0.05$, $\text{max_depth} = 6$, with L1/L2 regularisation

Gradient Boosting:

- $n_estimators = 300$, $\text{learning_rate} = 0.05$, $\text{max_depth} = 5$, $\text{subsample} = 0.8$

Appendix C – Project Files

- `dataset_generation.py` — synthetic agro-climatic dataset construction
- `preprocessing.py` — encoding, scaling, and temporal splitting
- `model_training.py` — training all six ML models
- `evaluation.py` — metric computation and result logging
- `feature_importance.py` — Random Forest feature importance extraction
- `visualisation.py` — scatter plots, residual histograms, and bar charts